COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE010	**SECTION:**	CPE21S1
DATE PERFORMED:	Aug 27, 2026	**DATE SUBMITTED:**	Aug 27, 2026
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ACTIVITY NO. __ [Prelim Laboratory Exam SET C]

1. PROCEDURE OUTPUT

Main.cpp

```cpp
#include <iostream>
#include <string>
#include "stackarray.h"
#include "queuelist.h"

bool checkPalindrome(std::string number){

    for(int i=0; i<(int)number.length(); i++){
        pushStack(number[i]);
        enqueue(number[i]);
    }

    bool answer=true;

    while(isEmptyQueue()==false){
        char fromStack=popStack();
        char fromQueue=dequeue();

        if(fromStack!=fromQueue)
            answer=false;
    }

    return answer;
}

int main(){

    std::string vals[5]={"10201", "1003003001", "88", "0", "202"};

    std::cout<<"Premade set of values\n";
    for(int i=0;i<5;i++){

        bool result=checkPalindrome(vals[i]);

        std::cout<<vals[i]<<" -> ";
        if(result==true){
            std::cout<<"Palindrome"<<std::endl;
        }
        else{
            std::cout<<"Not a Palindrome"<<std::endl;
        }
    }

    std::cout<<"\nPalindrome checker\n";
    std::string input;
    std::cout<<"enter number: ";
    std::cin>>input;

    bool res=checkPalindrome(input);
    if(res==true){
        std::cout<<input<<" is a Palindrome"<<std::endl;
    }
    else{
        std::cout<<input<<" is Not a Palindrome"<<std::endl;
    }

    return 0;
}
```

queuelist.h

```
#ifndef QUEUE_LIST_H
#define QUEUE_LIST_H

struct QNode{
    char data;
    QNode* next;
};

QNode* front=NULL;
QNode* rear=NULL;

bool isEmptyQueue(){
    return front==NULL;
}

void enqueue(char value){
    QNode* newNode=new QNode;
    newNode->data=value;
    newNode->next=NULL;

    if(rear==NULL){
        front=newNode;
        rear=newNode;
    }
    else{
        rear->next=newNode;
        rear=newNode;
    }
}

char dequeue(){
    if(isEmptyQueue()==false){
        QNode* temp=front;
        char value=temp->data;

        front=front->next;
        if(front==NULL)
            rear=NULL;

        delete temp;
        return value;
    }
    return '\0';
}

#endif
```

stackarray.h

```
#ifndef STACK_ARRAY_H
#define STACK_ARRAY_H

#include <iostream>

const int maxCap=100;
char stackArr[maxCap];
int top=-1;

bool isEmptyStack(){
    return top==-1;
}

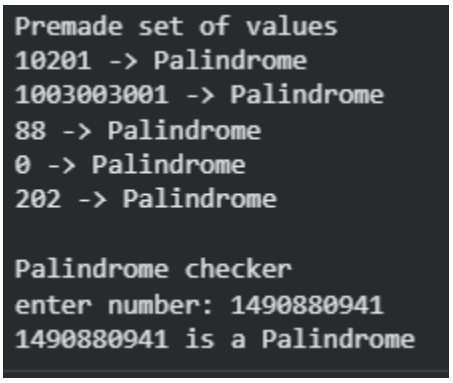
bool isFullStack(){
    return top==maxCap-1;
}

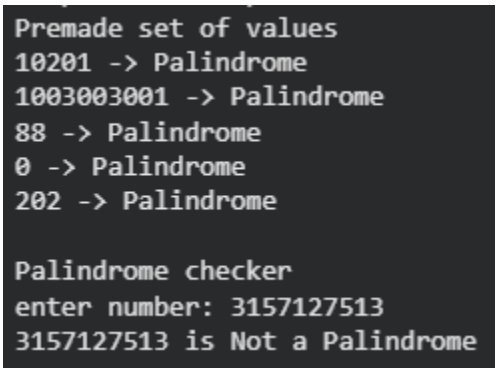
void pushStack(char value){
    if(isFullStack()==false){
        top++;
        stackArr[top]=value;
    }
}

char popStack(){
    if(isEmptyStack()==false){
        char value=stackArr[top];
        top--;
        return value;
    }
    return '\0';
}

#endif
```

Output





2. ANSWERS TO SUPPLEMENTAL ACTIVITY

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

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